Year 2 Statistics and Mechanics

Chapter 3

The normal distribution

Practice Booklet

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3.1 The normal distribution

Practice Book, printed page 21.

1. State, with a reason, whether or not the normal distribution might be suitable for modelling each of the following:

a. X, the number of children in a randomly chosen family

b. Y, the time taken for a particular employee to drive to work each day using the same route

c. R, the reaction time recorded by a person in an experiment

d. L, the length of a particular breed of earthworm selected at random in a particular area.

Hint

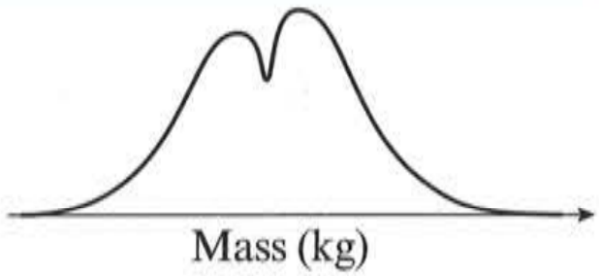
The normal distribution is used to model continuous data that take values grouped around a central value.

2. The scores on an IQ test, X, scored by a population of adults are normally distributed with mean 100 and standard deviation 15. Sketch the distribution of X.

Hint

The normal distribution curve is symmetrical about the mean and has points of inflection at $\mu + \sigma$ and $\mu - \sigma$. The horizontal axis is an asymptote at each end of the curve.

3. The masses, in kg, of a group of 200 people is shown.
State, with reasons, why the normal distribution is not a suitable model for this data.



4. The volume of soap dispensed by a soap-dispenser on each press, X ml, is modelled as $X \sim \mathrm{N}(7, 0.9^2)$.
State the proportion of presses that dispense:

a. between 6.1 ml and 7.9 ml

b. between 5.2 ml and 8.8 ml.

5. The masses, M kg, of a population of badgers are modelled as $M \sim \mathrm{N}(5, \sigma^2)$.
If 95% of the badgers have masses between 2.6 kg and 7.4 kg, find σ.

6. The weights of steel sheets produced by a factory are modelled as $W \sim \mathrm{N}(\mu, 2.2^2)$.
If 16% of steel sheets weigh more than 34.7 kg, find μ.

7. P The masses, Y grams, of a brand of chocolate bar are modelled as $Y \sim \mathrm{N}(\mu, \sigma^2)$.
If 84% of the chocolate bars weigh more than 40.5 g and 2.5% of the chocolate bars weigh more than 45 g, find μ and σ.

8. The thicknesses of table cloths, T, produced by a loom are modelled as a normal distribution with mean 2.5 mm and standard deviation 0.15 mm. Find:

a. $\mathrm{P}(T < 2.5)$

b. $\mathrm{P}(2.35 < T < 2.5)$

c. $\mathrm{P}(T > 2.8)$

9. P From the large data set, the daily mean temperatures, $T\,^\circ\mathrm{C}$, for Heathrow during October 1987 are shown below.

14.6	8.8	7.2	7.3	10.1	11.9	12.2
12.1	15.2	11.1	10.6	12.7	8.9	10.0
9.5	9.3	10.9	10.7	11.9	9.5	10.4
9.7	9.8	13.1	12.6	15.9	14.9	13.5
14.2	16.8	16.2				

a. Use the temperature data to find:

i. $\mathrm{P}(T < 11.6)$

ii. $\mathrm{P}(9 < T < 14.2)$

iii. $\mathrm{P}(6.5 < T < 16.8)$

The data has mean 11.6 and standard deviation 2.6.

b. Use your results from part **a** to decide whether the data can be modelled using the random variable $T \sim \mathrm{N}(11.6, 2.6^2)$.

3.2 Finding probabilities for normal distributions

Practice Book, printed page 22.

1. The random variable $X \sim \mathrm{N}(40, 4^2)$.
Find:

a. $\mathrm{P}(X < 43)$

b. $\mathrm{P}(X > 36)$

c. $\mathrm{P}(X > 38.6)$

Hint

Use the normal cumulative distribution function on your calculator to find probabilities from a normal distribution.

2. The random variable $X \sim \mathrm{N}(50, 25)$.
Find:

a. $\mathrm{P}(X > 58)$

b. $\mathrm{P}(X \leq 45)$

c. $\mathrm{P}(45 < X < 57)$

Hint

To find $\mathrm{P}(X > 58)$ on your calculator, choose an upper limit that is at least 5 standard deviations away from the mean, such as 100 or 500.

3. The random variable $Y \sim \mathrm{N}(36, 36)$.
Find:

a. $\mathrm{P}(Y < 30)$

b. $\mathrm{P}(30 < Y < 40)$

c. $\mathrm{P}(Y > 31.7)$

Hint

Make sure you enter the standard deviation into your calculator, and not the variance.

4. The random variable $X \sim \mathrm{N}(22, 8)$.
Find:

a. $\mathrm{P}(X > 20)$

b. $\mathrm{P}(X < 18)$

c. $\mathrm{P}(20.5 < X < 21.8)$

5. The random variable $M \sim \mathrm{N}(10, 1.2^2)$.

a. Find:

i. $\mathrm{P}(M > 11)$

ii. $\mathrm{P}(M \leq 11)$

b. Calculate the sum of your answers to parts **a i** and **ii** and comment on your answer.

6. The random variable $T \sim \mathrm{N}(5.2, 0.5)$.

a. Find $\mathrm{P}(T < 4.8)$.

b. Without further calculation, write down $\mathrm{P}(T > 4.8)$.

7. P The random variable $Y \sim \mathrm{N}(62, 5^2)$. Find:

a. $\mathrm{P}(Y < 58 \text{ or } Y > 65)$

b. $\mathrm{P}(Y < 60 \text{ or } 65 < Y < 70)$

8. E The lengths of screws, L mm, produced by a machine are modelled as $L \sim \mathrm{N}(100, 2.5^2)$.

a. Find:

i. $\mathrm{P}(X > 104)$

ii. $\mathrm{P}(X < 99)$

(2 marks)

Five screws are selected at random.

b. Find the probability that all of the screws are longer than 99 mm. (2 marks)

9. E/P A reading test is given to a group of year 2 students. The scores on the test are found to be normally distributed with $X \sim \mathrm{N}(80, 12^2)$.

a. Find:

i. $\mathrm{P}(X > 90)$

ii. $\mathrm{P}(X < 65)$

(2 marks)

A class of 25 year 2 students take the test. If three or more students score less than 60 then the class will have to retake the test.

b. Find the probability that the class will have to retake the test. (4 marks)

10. E/P The amount of jam, J grams, in each jar produced by a factory is modelled as $J \sim \mathrm{N}(454, 5.6^2)$. A jar is selected at random.

a. Find the probability that the jar contains more than 460 g of jam. (1 mark)

The jars pass a quality control test if they contain between 452 g and 458 g of jam.

b. Find the probability that a jar will pass the quality control test. (2 marks)

A sample of 18 jars is taken.

c. Find the probability that at least 15 pass the quality control test. (4 marks)

11. E/P The heights of a group of female athletes are modelled by a normal distribution with mean 165 cm and standard deviation 4.8 cm. The weights of this group of athletes are modelled by a normal distribution with mean 65 kg and standard deviation 6.7 kg.

a. Assuming that for these athletes height and weight are independent, find the probability that a randomly chosen athlete is taller than 170 cm and weighs more than 68 kg. (4 marks)

b. Comment on the assumption that height and weight are independent. (1 mark)

3.3 The inverse normal distribution function

Practice Book, printed page 24.

1. The random variable $X \sim \mathrm{N}(40, 3^2)$. Find the value of a, to 2 decimal places, such that:

a. $\mathrm{P}(X < a) = 0.25$

b. $\mathrm{P}(X < a) = 0.75$

c. $\mathrm{P}(X > a) = 0.4$

Hint

Use the inverse normal distribution function on your calculator. In part **c**, use that fact that $\mathrm{P}(X > a) + \mathrm{P}(X < a) = 1$ to find the area to the left of a before using your calculator.

2. The random variable $X \sim \mathrm{N}(15, 2^2)$. Find the value of a, to 2 decimal places, such that:

a. $\mathrm{P}(X < a) = 0.2$

b. $\mathrm{P}(X > a) = 0.75$

c. $\mathrm{P}(13 \leq X \leq a) = 0.4$

d. $\mathrm{P}(a < X < 18) = 0.43$

Hint

Use the inverse normal distribution function on your calculator.
In part **c**, use that fact that $\mathrm{P}(13 \leq X \leq a) = \mathrm{P}(X \leq a) - \mathrm{P}(X < 13)$

3. The random variable $X \sim \mathrm{N}(25, 15)$.

a. Find the value of a and the value of b such that:

i. $\mathrm{P}(X < a) = 0.303$

ii. $\mathrm{P}(X > b) = 0.303$

b. Find $\mathrm{P}(b < X < a)$.

4. The random variable $Y \sim \mathrm{N}(80, 25)$.

a. Find the value of a and the value of b such that:

i. $\mathrm{P}(Y < a) = 0.212$

ii. $\mathrm{P}(Y < b) = 0.885$

b. Find $\mathrm{P}(a < Y < b)$.

5. P The random variable $X \sim \mathrm{N}(100, 36)$.

a. Find values of a and b such that $\mathrm{P}(a < X < b) = 0.6$.

b. Explain briefly why there are other values of a and b with this property.

6. The distances travelled to work, D km, by the employees at a large company are normally distributed with $D \sim \mathrm{N}(25, 7^2)$.
For this population, find:

a. the upper quartile

b. the 30th percentile

c. the value of Q_2.

7. E A packing plant fills bags with cement. The weight X kg of a bag of cement can be modelled by a normal distribution with mean 60 kg and standard deviation 3 kg.

a. Find the value of a such that $P(X > a) = 0.4$. (1 mark)

b. Find the interquartile range of the scores. (2 marks)

8. E/P The weights of bags of popcorn, W g, produced by a factory are normally distributed with $W \sim N(175, 10^2)$.

a. Find the value of b such that $P(Y < b) = 0.85$. (1 mark)

b. Find the 20% to 80% interpercentile range of masses. (2 marks)

9. E/P The heights of seedlings are normally distributed with mean 100 mm and standard deviation 6.4 mm.
The probability that a randomly selected seedling is at least $(100 + k)$ mm high is 0.1379.
Find, to the nearest integer, the value of k. (2 marks)

3.4 The standard normal distribution

Practice Book, printed page 25.

1. For the standard normal distribution $Z \sim N(0, 1^2)$, find:

a. $P(Z < 1.88)$

b. $P(Z > 1.25)$

c. $P(Z < -0.77)$

d. $P(-1.64 < Z < 0)$

e. $P(-2.55 < Z < -0.78)$

f. $P(-1.15 < Z < 1.15)$

Hint

The standard normal distribution has mean 0 and standard deviation 1.

2. For the standard normal distribution $Z \sim N(0, 1^2)$, find values of a such that:

a. $P(Z < a) = 0.9177$

b. $P(Z > a) = 0.0351$

c. $P(Z > a) = 0.8315$

d. $P(0 < Z < a) = 0.1700$

e. $P(0 < Z < a) = 0.4970$

f. $P(-a < Z < a) = 0.60$

Hint

For part **f**, use the fact that the normal distribution is symmetrical about its mean.

3. The random variable $X \sim N(1.4, 0.25^2)$. For each of the following values of X, write down the corresponding value of the standardised normal distribution, $Z \sim N(0, 1^2)$.

a. $x = 1.2$

b. $x = 0.625$

c. $x = 0.64$

d. $x = 1.981$

Hint

If $X \sim N(\mu, \sigma^2)$, and the values of X are coded using $Z = \dfrac{X - \mu}{\sigma}$, then $Z \sim N(0, 1^2)$.

4. The normal distribution $X \sim N(124, 10^2)$. Write in terms of $\Phi(z)$:

a. $P(X < 136)$

b. $P(X > 140)$

c. $P(120 < X < 132)$

5. E

a. Use the percentage points table to find a value of z such that $P(Z > z) = 0.05$. (1 mark)

A university offers academic awards to the top 5% of students taking an examination.

b. Given that the scores can be modelled using a normal distribution with mean 65 and standard deviation 7.2, use your answer to part **a** to find the score necessary to receive an award. (2 marks)

6. E

a. Use the percentage points table to find a value of z such that $P(Z < z) = 0.25$. (1 mark)

An electronics store offers a 'basic' insurance policy for laptops. The policy should be suitable for 12.5% of the lowest priced laptops sold in the store.

b. Given that the prices of laptops can be modelled using a normal distribution with mean £177 and standard deviation £33, use your answer to part **a** to find the maximum price of a laptop for which the 'basic' insurance policy is suitable. (2 marks)

7. E

a. Use the percentage points table to find the values of z that correspond to the 20% to 80% interpercentile range. (2 marks)

A pottery student makes plates that are intended to be 20 cm in diameter. The actual diameter of each finished plate, D cm, can be modelled as $D \sim N(20, 1.5^2)$. The pottery student will be able to sell any plates that fall in the 20% to 80% interpercentile range.

b. Use your answer to part **a** to find the largest and smallest diameters of plates which the student will be able to sell. (2 marks)

3.5 Finding μ and σ

Practice Book, printed page 26.

1. The random variable $X \sim N(\mu, 4^2)$ and $P(X < 26) = 0.7734$.
Find the value of μ.

Hint

You need to use the standard normal distribution.
Find the value of z such that $P(Z < z) = 0.7734$
Then use the coding formula $z = \dfrac{X - \mu}{\sigma}$ to find μ.

2. The random variable $X \sim N(17, \sigma^2)$ and $P(X > 20) = 0.1151$.
Find the value of σ.

3. The random variable $Y \sim N(\mu, 30)$ and $P(Y < 28) = 0.0721$.

a. Find the value of μ.

b. Find $P(Y < 40)$

4. The random variable $Y \sim N(50, \sigma^2)$ and $P(Y > 40) = 0.6554$.

a. Find the value of σ.

b. Find $P(45 < Y < 50)$

5. P The random variable $X \sim N(\mu, \sigma^2)$.
Given that $P(X < 25) = 0.9401$ and $P(X < 22) = 0.8130$, find the values of μ and σ.

Hint

Use the two probabilities given to form two equations involving μ and σ, then solve these equations simultaneously.

6. P The random variable $Y \sim N(\mu, \sigma^2)$.
Given that $P(Y < 5) = 0.1817$ and $P(Y > 11) = 0.0345$, find:

a. the value of μ and the value of σ

b. $P(4 < Y < 8)$

7. E/P An angler is investigating the lengths, X cm, of trout caught at a particular location. She knows that the lower quartile of the lengths is 24.9 cm and the upper quartile of the lengths is 26.1 cm. She assumes that the lengths will be normally distributed.

a. Give two reasons why this assumption might be valid. (2 marks)

b. Find the mean and standard deviation of the distribution of lengths. (3 marks)

c. Find the probability that a trout caught at random has a length between 24 cm and 25 cm. (1 mark)

8. P The random variable $X \sim \mathrm{N}(0, \sigma^2)$.
Given that $\mathrm{P}(-5 < X < 5) = 0.75$, find the value of σ.

9. E/P The random variable $X \sim \mathrm{N}(1.45, \sigma^2)$.
Given that $\mathrm{P}(X > 2a) = 0.3$ and $\mathrm{P}(X < a) = 0.6$, find the value of σ and the value of a. (6 marks)

10. E/P The scores, X points, of a group of learner drivers in the hazard perception section of their theory test is modelled as a normal distribution with mean μ and standard deviation 7.2. A learner driver needs to score more than 44 points to pass the hazard perception test. Given that 39% of learner drivers pass the hazard perception test, find:

a. the value of μ (2 marks)

b. $\mathrm{P}(48 < X < 53)$. (1 mark)

Three learner drivers are chosen at random.

c. Find the probability that all of the drivers score more than 50 points. (3 marks)

11. E/P The diameters of bolts, D mm, made by a particular machine are modelled using a normal distribution with mean 12.5 mm. Given that 80% of bolts have a diameter less than 13 mm, find:

a. the standard deviation of the diameter (2 marks)

b. the proportion of bolts with diameters between 12.4 mm and 12.6 mm. (1 mark)

A bolt is accepted if its diameter is between 12.3 mm and 12.7 mm.
A sample of 10 bolts is taken.

c. Find the probability that at least 5 bolts will be accepted. (3 marks)

12. E/P The masses of adult squirrels are found to be normally distributed with mean μ and standard deviation σ. Given that 1 in 3 squirrels have a mass less than 455 g and 1 in 5 have a mass greater than 500 g,

a. sketch a diagram to represent this information (2 marks)

b. find the value of μ and the value of σ (6 marks)

c. find the 10th to 90th percentile range. (2 marks)

3.6 Approximating a binomial distribution

Practice Book, printed page 27.

1. For each of the following binomial random variables, X:

i. state, with reasons, whether X can be approximated by a normal distribution

ii. if appropriate, write down the normal approximation to X in the form $\mathrm{N}(\mu, \sigma^2)$, giving the values of μ and σ.

a. $X \sim \mathrm{B}(250, 0.9)$

b. $X \sim \mathrm{B}(12, 0.48)$

c. $X \sim \mathrm{B}(120, 0.52)$

d. $X \sim \mathrm{B}(420, 0.45)$

e. $X \sim \mathrm{B}(600, 0.54)$

f. $X \sim \mathrm{B}(800, 0.74)$

Hint

When n is large and p is close to 0.5, you can approximate a binomial distribution $\mathrm{B}(n, p)$ with a normal distribution $\mathrm{N}(np, np(1-p))$.

2. The random variable $X \sim \mathrm{B}(300, 0.47)$.
Use a suitable approximation to estimate:

a. $\mathrm{P}(X \leq 140)$

b. $\mathrm{P}(120 \leq X \leq 160)$

c. $\mathrm{P}(X = 150)$

Hint

X is a discrete distribution so you need to use a continuity correction. If the binomial random variable X is approximated with the normal random variable Y, then $\mathrm{P}(X \leq 140) \approx \mathrm{P}(Y < 140.5)$

3. The random variable $X \sim \mathrm{B}(250, 0.52)$. Use a suitable approximation to estimate:

a. $\mathrm{P}(X > 120)$

b. $\mathrm{P}(115 < X < 150)$

c. $\mathrm{P}(X = 125)$

4. A fair dice is thrown 125 times. Use a suitable approximation to estimate the probability of obtaining more than 60 even scores.

5. E Students taking an intensive driving course pass their driving test with a 56% success rate.
A random sample of 15 students is taken and X pass their driving test.

a. Find $\mathrm{P}(X = 10)$ (1 mark)

A second random sample of 250 students is taken.

b. Using a suitable approximation, find the probability that fewer than 130 students in the sample will pass their driving test. (3 marks)

c. The probability that at least q students pass their test is 0.1. Find q. (3 marks)

6. E It is known that 47% of patients calling a GP surgery are given a next day appointment.
A random sample of 80 patients is taken.

a. Find the probability that exactly 40 patients are given a next day appointment. (1 mark)

b. Find the percentage error when using a normal approximation to calculate the probability that exactly 40 patients are given a next day appointment. (4 marks)

7. E/P It is suggested to a student that the random variable $X \sim \mathrm{B}(220, 0.55)$ can be approximated by a normal random variable Y.
The student is asked to use this approximation to find an estimate for $\mathrm{P}(X \leq 110)$.
The student's attempt is shown here.

> $Y \sim \mathrm{N}(\mu, \sigma^2)$
> $\mu = np = 220 \times 0.55 = 121$
> $\sigma = \sqrt{np(1-p)} = \sqrt{220 \times 0.55 \times (1 - 0.55)} = \sqrt{54.45}$
> $Y \sim (121, 54.45)$
> $\mathrm{P}(X \leq 110) \approx \mathrm{P}(Y \leq 110) = 0.068$

a. Identify the error made by the student. (1 mark)
b. Find the correct estimate. (1 mark)

Use the normal approximation to the binomial distribution to find estimates for:
c. $\mathrm{P}(X > 130)$ (1 mark)
d. $\mathrm{P}(105 < X < 120)$ (1 mark)

3.7 Hypothesis testing with the normal distribution

Practice Book, printed page 29.

1. The distances, X metres, that Michael can hit a golf ball are normally distributed with mean 200 and standard deviation 15. Michael wants to test, at the 5% significance level, whether a new golf club has increased the distance that he can hit a golf ball.
Michael uses the new club 50 times and records the distances that he hits the ball.
a. Write down a suitable test statistic.
b. Write down two suitable hypotheses.
c. Write down the distribution of the sample mean.
d. Find the critical region for the test statistic.

Hint

The sample mean has distribution $\overline{X} \sim \mathrm{N}\left(\mu, \dfrac{\sigma^2}{n}\right)$

2. A sample of size 30 is taken from a normal distribution $X \sim \mathrm{N}(\mu, 5.7^2)$ and is found to have a sample mean of $\bar{x} = 16.6$.
The sample is used to test $\mathrm{H}_0 : \mu = 18$ against $\mathrm{H}_1 : \mu < 18$ at the 10% level of significance.
a. Explain the condition under which the null hypothesis would be rejected.
b. Find $\mathrm{P}(\overline{X} < 16.6)$ and state the result of the hypothesis test.

3. The times a mobile phone battery lasts before needing to be recharged are assumed to be normally distributed with a mean of 45 hours and a standard deviation of 6 hours.
The manufacturers of the mobile phone claim to have developed a new battery with a longer battery life. The manufacturers test a sample of 50 batteries to determine whether this is true.
a. Find, at the 5% level, the critical region for this test, stating your hypotheses clearly.
The mean battery life for the sample of 50 batteries was found to be 50 hours.
b. Comment on this observation in light of the critical region.

Hint

You may assume that the standard deviation of the battery lives is unchanged.

4. E

a. Explain the difference between a one-tailed hypothesis test and a two-tailed hypothesis test for the mean of a normal distribution. (1 mark)

A sample of size 25 is taken from a normal distribution $X \sim \mathrm{N}(\mu, 1.1^2)$, and is used to test $\mathrm{H}_0 : \mu = 34$ against $\mathrm{H}_1 : \mu \neq 34$ at the 1% significance level.

b. Find the critical region for the test statistic and state the result of the hypothesis test. (3 marks)

The sample is found to have a mean of $\bar{x} = 33.7$.

c. Comment on this observation in light of your answer to part **b**. (1 mark)

5. E An automated pottery wheel is used to make bowls. The diameters of the bowls, D mm, are normally distributed with mean 200 mm and standard deviation 6 mm. After the wheel is repaired, a random sample of 20 bowls is taken and their diameters are measured to see if the mean diameter has altered.

The mean of the sample is found to be 201.5 mm.

Test, at the 10% significance level, whether there is evidence of a change in the mean diameter of bowls produced by the automated pottery wheel. (4 marks)

6. E The masses of gentoo penguins on an island are found to be normally distributed with mean 5.9 kg and standard deviation 0.84 kg. A zoologist believes gentoo penguins in captivity have larger masses.

A random sample of 16 gentoo penguins are chosen from a zoo. The mean mass of the sample is found to be 6.6 kg.

Test, at the 1% level, whether this sample provides sufficient evidence to support the zoologist's claim. (4 marks)

7. E/P The waiting times at a doctor's surgery are assumed to be normally distributed with mean 12 minutes and standard deviation 3.6 minutes.

A new patient check-in system is introduced to reduce waiting times. After the system is introduced the surgery manager wants to determine if the waiting times have been reduced.

The surgery manager takes a sample of 25 patient times.

a. State suitable null and alternative hypotheses to test whether the new check-in system has reduced waiting times. (1 mark)

b. Determine the maximum mean waiting time the surgery manager could observe in order to reject H_0 at the 1% significance level. (4 marks)

In fact, the surgery manager decides to test at the 10% significance level and observes a mean waiting time for the sample of 12.3 minutes.

c. Explain briefly why the surgery manager can draw a conclusion without making any further calculations, and state this conclusion. (2 marks)

Problem solving: Set A

Bronze

From experience, a heptathlete knows that she is able to throw her javelin a distance of 45 m or more once in every 5 attempts. Given that the distances, X m, that the javelin is thrown can be assumed to be normally distributed with $X \sim \mathrm{N}(40, \sigma^2)$,

a. draw a sketch to show the distribution of X, labelling the area of the region represented by $X > 45$ (2 marks)

b. find, to 2 decimal places, the value of σ. (2 marks)

The heptathlete throws her javelin.

c. Calculate the probability that she throws it further than 42 m. (2 marks)

Silver

A machine cuts strips of metal to length L cm, where L is normally distributed with $L \sim \mathrm{N}(\mu, \sigma^2)$. Given that 20% of the cut lengths are less than 19.50 cm and 5% of the cut lengths are greater than 20.76 cm,

a. find the value of μ and the value of σ. (6 marks)

Strips with lengths between 19.75 cm and 20.5 cm are graded as 'firsts', and can be sold for a higher price.

b. Find the proportion of strips produced which can be graded as 'firsts'. (2 marks)

Gold

In a 'test-your-strength' game at a fairground, the players must hit a button with a hammer to make a pointer rise up a 2-metre column. The game operator knows that players win no prize with probability $\frac{1}{5}$ and a small prize with probability $\frac{3}{4}$. The operator assumes that the heights, X cm, reached by players can be modelled using a normal distribution.

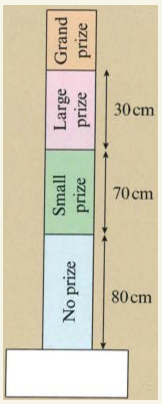
a. Using this assumption, find the mean and standard deviation of this distribution. (3 marks)

On one day, 300 people play the game.

b. Estimate the number of large prizes the fairground owner can expect to have to give out. (2 marks)

The fairground owner states that a normal distribution is not a suitable model as in reality it is not possible for players to reach heights above 2 m.

c. With reference to your answer to part **a**, state, with justification, whether you agree with the fairground owner's statement. (2 marks)



Problem solving: Set B

Bronze

Applicants for a job are given an aptitude test. The percentage, X%, achieved by a randomly chosen applicant is modelled as $X \sim \mathrm{N}(50, 10^2)$. Applicants are scored as 'outstanding' if they achieve 65% or more. A group of 8 applicants are assessed on one day.

a. Find the probability that:

i. at least one is scored as outstanding

ii. exactly two are scored as outstanding

iii. fewer than three are scored as outstanding.

(6 marks)

An interviewer claims that the model is unsuitable as, according to the model, an applicant could score more than 100%.

b. Evaluate the interviewer's claim. (1 mark)

Silver

The scores, X points, of a group of learner drivers in the theory test is modelled as a normal distribution with mean 38 and standard deviation 6.4. A learner driver needs to score more than 43 points to pass the test.

a. Find: i $\mathrm{P}(X > 43)$ ii $\mathrm{P}(X > 45)$ (2 marks)

b. Explain briefly why $\mathrm{P}(X > 43 \text{ and } X > 45) = \mathrm{P}(X > 45)$. (1 mark)

A learner driver is chosen at random.

c. Given that the driver passed the test, find the probability that the driver scored more than 45 points. (2 marks)

Gold

Anton is modelling the lifetime of low energy lightbulbs in his house. He finds that with normal usage, the lifetime, X days, of a particular type of bulb can be modelled as $X \sim \mathrm{N}(720, 150^2)$. Anton's dining room light fixture uses 6 of these bulbs. Anton installs 6 new bulbs in this light fixture.

a. Find the probability that at least 5 of them will last for more than 700 days. (4 marks)

After 400 days, none of the bulbs has failed.

b. Find the probability that all 6 bulbs will continue to last for at least another 300 days. (5 marks)

After these 400 days, Anton decides to replace 4 of the bulbs at random with new bulbs.

c. Find the new probability that all 6 bulbs will last for at least another 300 days. (3 marks)

Selected exam question bank questions

Question 3

The random variable X is normally distributed with mean μ and variance σ^2.

a. Write down the distribution of the sample mean $\overline{X}$ of a random sample of size n. (1 mark)

A machine in a factory makes metal sheets with width X cm. The factory supervisor wants to use a sample to estimate the mean of X.

b. Determine how large a random sample is needed so that the supervisor can be 95% certain that the sample mean width will differ from the population mean width by less than 1 cm. Assume that it is known that $\sigma = 3.2$ cm. (4 marks)

Question 6

a. Write down two conditions under which the normal distribution may be used as an approximation to the binomial distribution. (2 marks)

A student is given that the random variable $X \sim \mathrm{B}(250, 0.48)$ can be approximated by a normal random variable Y.

The student is asked to find an estimate for $\mathrm{P}(X < 130)$.

The student's attempt is shown

$Y \sim \mathrm{N}(\mu, \sigma^2)$
$\mu = np = 250 \times 0.48 = 120$
$\sigma = np(1-p) = 250 \times 0.48 \times 0.52 = 62.4$
$Y \sim (120, 62.4^2)$
$\mathrm{P}(X < 130) \approx \mathrm{P}(Y < 129.5) = 0.5605$

b. Identify the error made by the student. (1 mark)

c. Use a normal approximation to find the correct estimate for $\mathrm{P}(X < 130)$. (2 marks)

Question 8

In a probability experiment, two fair six-sided dice are thrown. A trial is considered a 'success' if the larger number shown on either dice is prime.

a. Find the probability of success on a single trial. (3 marks)

The experiment is repeated 180 times, and the number of successes, X, is recorded.

b. Write down the distribution of X. (1 mark)

c. Explain why X can be approximated with a normal distribution $X \sim \mathrm{N}(\mu, \sigma^2)$, giving the values of μ and σ. (4 marks)

d. Use this normal distribution to estimate the probability that $X = 80$. (1 mark)

Question 11

A packing plant fills bags with gravel. The weights, X kg, of bags of gravel can be modelled by a normal distribution with mean 40 kg and standard deviation 2 kg.

a. Find $P(X > 42)$. (2 marks)

b. Find the weight that is exceeded by 1% of the bags. (3 marks)

Three bags are selected at random.

c. Find the probability that two weigh more than 41 kg and one weighs less than 41 kg. (4 marks)

Bags are rejected if they weigh less than 40 kg.

d. Given that a bag is not rejected, find the probability that it weighs more than 42 kg. (3 marks)

Question 13

The amount of mineral water, W ml, in a bottle produced by a certain manufacturer is found to be normally distributed with mean μ and standard deviation σ.

Given that 78% of bottles contain more than 320 ml of water and 38% contain more than 334 ml of water,

a. find the value of μ and the value of σ. (6 marks)

The bottles are packed in boxes of 8 bottles. A box meets quality standards if no more than 2 bottles contain less than 325 ml of water.

b. Find the probability that a randomly chosen box meets quality standards. (3 marks)

A supermarket orders 600 boxes of mineral water.

c. Using a suitable approximation, find the probability that fewer than 250 of these boxes meet quality standards. (4 marks)

Question 15

A call-centre collected data on how long customers had to wait before an operator was free to take their call.

The lower quartile of the data was 12.7 minutes and the interquartile range was 6.2 minutes.

It is suggested that a normal distribution could be used to model the waiting times.

a. Using this model, calculate:

i. the mean and standard deviation of the waiting times, correct to 3 s.f.

ii. the probability that a randomly chosen customer will have to wait longer than 16 minutes. (7 marks)

A customer has already been waiting for 12 minutes.

b. Find the probability that this customer will have to wait more than an additional 8 minutes. (3 marks)

On one day, two calls took more than three quarters of an hour to be answered.

c. Comment on the suitability of the model in light of this information. (1 mark)

Question 19

The reading scores, X, of a population of 7-year-olds are normally distributed with a mean of 50 and a standard deviation of 8.

a. Find $\mathrm{P}(X > 55)$. (2 marks)

b. Find the upper quartile Q_3 of X. (3 marks)

c. Write down the lower quartile Q_1 of X. (1 mark)

An outlier is defined as any value of X such that $X < h$ or $X > k$ where

$$h = Q_1 - 1.5(Q_3 - Q_1) \quad \text{and} \quad k = Q_3 + 1.5(Q_3 - Q_1)$$

d. Find the value of h and the value of k. (2 marks)

A child is selected at random.

e. Find the probability that their reading score is an outlier. (3 marks)

A teacher wants to test the theory that reading for an extra 20 minutes a day will improve reading scores. A random sample of 12 children are chosen and instructed to read for an extra 20 minutes a day for a month. They are then tested to find their reading scores.

The mean score of the sample was found to be 54.

f. Stating your hypotheses clearly, and using a 5% level of significance, test whether reading for 20 minutes a day has improved reading scores. (4 marks)

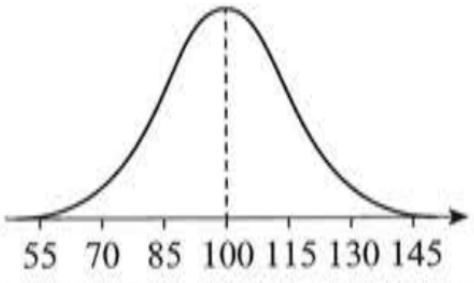
g. Give one criticism of your conclusion to part **f**. (1 mark)

Answers

3.1 The normal distribution

1.
- **a.** Not suitable: number of children is discrete.
- **b.** Not suitable: the distribution is unlikely to be symmetrical.
- **c.** Suitable, e.g. likely to be similar times.
- **d.** Suitable, e.g. likely to be similar length.

2.



3. The distribution shown does not have a bell-shaped curve.

4.
- **a.** 68.2% (3 s.f.)
- **b.** 95.4% (3 s.f.)

5. 1.2

6. 32.5 kg

7. $\mu = 42$ g, $\sigma = 1.5$ g

8.
- **a.** 0.5
- **b.** 0.341 (3 s.f.)
- **c.** 0.0228 (3 s.f.)

9.
- **a.**
  - **i.** 0.516 (3 s.f.)
  - **ii.** 0.613 (3 s.f.)
  - **iii.** 1
- **b.** Using the normal distribution the probabilities are 0.5, 0.68 and 0.95. The probabilities are close, so the model is suitable.

3.2 Finding probabilities for normal distributions

1.
- **a.** 0.7734
- **b.** 0.8413
- **c.** 0.6368

2.
- **a.** 0.0548
- **b.** 0.1587
- **c.** 0.7606

3.
- **a.** 0.1587
- **b.** 0.5889
- **c.** 0.7632

4.
- **a.** 0.7602
- **b.** 0.0786
- **c.** 0.1739

5.
- **a.**
  - **i.** 0.2023
  - **ii.** 0.7977
- **b.** $P(M > 11) + P(M < 11) = 1$ which is the total probability of all possible events, so $P(M = 11) = 0$.

6.
- **a.** 0.2858
- **b.** 0.7142

7.
- **a.** 0.4861
- **b.** 0.5641

8.
- **a.**
  - **i.** 0.0548
  - **ii.** 0.3446
- **b.** 0.1209

9.
- **a.**
  - **i.** 0.2023
  - **ii.** 0.1056
- **b.** 0.1150

10.
- **a.** 0.1420
- **b.** 0.4020
- **c.** 0.0002

11.
- **a.** 0.0487
- **b.** The height of an athlete is likely to have an effect on the athlete's weight, so the assumption of independence is not sensible.

3.3 The inverse normal distribution function

1.
- **a.** 37.98
- **b.** 42.02
- **c.** 40.76

2.
- **a.** 13.32
- **b.** 13.65
- **c.** 15.30
- **d.** 15.02

3.
- **a.**
  - **i.** 23.00
  - **ii.** 27.00
- **b.** 0.394

4.
- **a.**
  - **i.** 76.0 (3 s.f.)
  - **ii.** 86.0 (3 s.f.)
- **b.** 0.673

5.
- **a.** E.g. $a = 94.9$, $b = 105$ (3 s.f.)
- **b.** For any value of a chosen there is a value of b such that $\mathrm{P}(a < X < b) = 0.6$.

6.
- **a.** 29.7 (3 s.f.)
- **b.** 21.3 (3 s.f.)
- **c.** 25

7.
- **a.** 60.8 (3 s.f.)
- **b.** 4.05 (3 s.f.)

8.
- **a.** 185 (3 s.f.)
- **b.** 16.8 (3 s.f.)

9. $k = 7$

3.4 The standard normal distribution

1.
- **a.** 0.9699
- **b.** 0.1056
- **c.** 0.2206
- **d.** 0.4495
- **e.** 0.2123
- **f.** 0.7499

2.
- **a.** 1.39 (3 s.f.)
- **b.** 1.81 (3 s.f.)
- **c.** -0.960 (3 s.f.)
- **d.** 0.440 (3 s.f.)
- **e.** 2.75 (3 s.f.)
- **f.** 0.841 (3 s.f.)

3.
- **a.** -0.8
- **b.** -3.1
- **c.** -3.04
- **d.** 2.324

4.
- **a.** $\Phi(1.2)$
- **b.** $1 - \Phi(1.6)$
- **c.** $\Phi(0.8) + \Phi(0.4) - 1$

5.
- **a.** 1.645 (4 s.f.)
- **b.** 77 (nearest whole number)

6.
- **a.** -0.675 (3 s.f.)
- **b.** £155

7.
- **a.** -0.8416, 0.8416 (4 s.f.)
- **b.** 21.26 cm, 18.74 cm (4 s.f.)

3.5 Finding μ and σ

1. 23.0 (3 s.f.)

2. 2.50 (3 s.f.)

3.
- **a.** 36.0 (3 s.f.)
- **b.** 0.7674

4.
- **a.** 25.0 (3 s.f.)
- **b.** 0.0793

5. $\mu = 18$, $\sigma = 4.5$

6.
- **a.** $\mu = 7.00$ (3 s.f.), $\sigma = 2.20$ (3 s.f.)
- **b.** 0.5889

7.
- **a.** The lengths tend to be symmetrical around a mean. The distribution of lengths will form a bell curve.
- **b.** Mean = 25.5 cm, standard deviation = 0.890 (3 s.f.)
- **c.** 0.2412

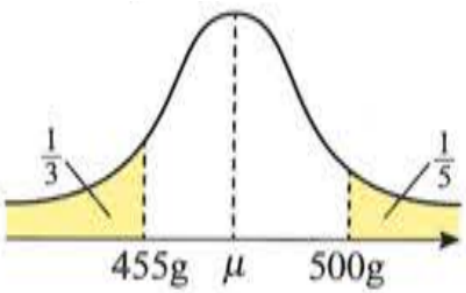
8. 4.35 (3 s.f.)

9. $\sigma = 81.9$ (3 s.f.), $a = 22.2$ (3 s.f.)

10.
- **a.** 42.0 (3 s.f.)
- **b.** 0.1390
- **c.** 0.0027

11.
- **a.** 0.5941 (4 s.f.)
- **b.** 0.1337
- **c.** 0.0950

12.
- **a.**



- **b.** $\mu = 470$ (3 s.f.), $\sigma = 35.4$ (3 s.f.)
- **c.** 90.7 (3 s.f.)

3.6 Approximating a binomial distribution

1.

a.

i. Not suitable, p not close to 0.5

b.

i. Not suitable, n not large

c.

i. Suitable, n large and p close to 0.5

ii. $\mathrm{N}(62.4, 5.47^2)$ (3 s.f.)

d.

i. Suitable, n large and p close to 0.5

ii. $\mathrm{N}(189, 10.2^2)$ (3 s.f.)

e.

i. Suitable, n large and p close to 0.5

ii. $\mathrm{N}(324, 12.2^2)$

f.

i. Not suitable, p not close to 0.5

2.

a. 0.4769

b. 0.9815

c. 0.0268

3.

a. 0.8854

b. 0.9600

c. 0.0413

4. 0.6397

5.

a. 0.1502

b. 0.0905

c. 150

6.

a. 0.0770

b. 0.3% (1 s.f.)

7.

a. The student did not apply the continuity correction
$\mathrm{P}(X \leq 110) \approx \mathrm{P}(Y \leq 110.5)$.

b. 0.0774

c. 0.0990

d. 0.4016

3.7 Hypothesis testing with the normal distribution

1.

a. The mean of the 50 distances in the sample (or $\bar{x}$, or the sample mean)

b. $\mathrm{H}_0 : \mu = 200$, $\mathrm{H}_1 : \mu > 200$

c. $\overline{X} \sim \mathrm{N}\left(200, \dfrac{15^2}{50}\right)$

d. $\overline{X} > 203.5$ (4 s.f.)

2.

a. Rejected if $\mathrm{P}(\overline{X} \leq \bar{x}) < 0.1$.

b. $\mathrm{P}(\overline{X} < 16.6) = 0.0893$ (4 s.f.). Reject H_0, accept H_1.

3.

a. $H_0 : \mu = 45$, $H_1 : \mu > 45$, $\overline{X} > 46.4$ hours

b. $\bar{x} = 50 > 46.4$
There is evidence that the new battery is an improvement.

4.

a. A one-tailed test is used to test when it is claimed that the mean has increased or decreased. A two-tailed test is used when it is thought that the mean has changed in either direction.

b. $\overline{X} < 33.4$ or $\overline{X} > 34.6$ (3 s.f.)

c. 33.7 is not in the critical region, so accept H_0.

5. $H_0 : \mu = 200$, $H_1 : \mu \neq 200$.
Critical region is $\overline{X} < 197.8$, $\overline{X} > 202.2$ (4 s.f.). Accept H_0.

6. $H_0 : \mu = 5.9$, $H_1 : \mu > 5.9$. Critical region is $\overline{X} > 6.389$ (4 s.f.).
Reject H_0, there is evidence for the zoologist's claim.

7.

a. $H_0 : \mu = 12$, $H_1 : \mu < 12$

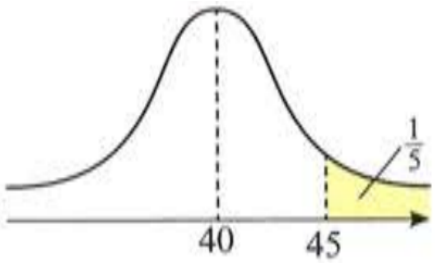
b. 10.3 minutes (3 s.f.)

c. $12.3 > 12$ so $P(\overline{X} < \bar{x})$ will be greater than 50%. There is insufficient evidence to reject H_0, so conclude the waiting times have not reduced.

Problem solving: Set A

Bronze

a.



b. $\sigma = 5.94$

c. 0.3682

Silver

a. $\mu = 19.9$ (3 s.f.), $\sigma = 0.507$ (3 s.f.)

b. 50.9% (3 s.f.)

Gold

a. $\mu = 104$ cm (3 s.f.), $\sigma = 28.2$ cm (3 s.f.)

b. 14 prizes

c. 200 cm is more than 3 standard deviations from the mean, so according to the model the probability of getting a value > 200 cm is very small (less than 0.001) so the model is still suitable.

Problem solving: Set B

Bronze

a.

i. 0.4247

ii. 0.0825

iii. 0.9871

b. 100% is 5 standard deviations from the mean so virtually impossible to occur as an outcome, so the interviewer's claim is not valid.

Silver

a.

i. 0.2173

ii. 0.1370

b. If $X > 45$ then X must be greater than 43, so for both to be true we only need $X > 45$.

c. 0.6305

Gold

a. 0.1673

b. 0.0316

c. 0.3128

Selected exam question bank answers

Question 3

a. $\overline{X} \sim \text{N}\left(\mu, \frac{\sigma^2}{n}\right)$

b. 21

Question 6

a. p is close to 0.5; n is large

b. Incorrect calculation of standard deviation, $\sigma = \sqrt{62.4}$

c. 0.8854

Question 8

a. $\frac{17}{36}$

b. $X \sim \text{B}\left(180, \frac{17}{36}\right)$

c. n is large and p (0.4722) is close to 0.5, $X \sim \text{N}(85, 6.7^2)$.

d. 0.0451 (3 s.f.)

Question 11

a. 0.1587

b. 44.65 kg (4 s.f.)

c. 0.197

d. 0.3173

Question 13

a. $\mu = 330$ ml (3 s.f.), $\sigma = 13.0$ (3 s.f.)

b. 0.430 (3 s.f.)

c. 0.248 (3 s.f.)

Question 15

a.

i. $\mu = 15.8$, $\sigma = 4.60$

ii. 0.483 (3 s.f.)

b. 0.227 (3 s.f.)

c. 45 mins is less than 6 standard deviations away from the mean, so virtually impossible according to the normal distribution model. The model is unlikely to be suitable.

Question 19

a. 0.2660

b. 55.4 (3 s.f.)

c. 44.6 (3 s.f.)

d. $h = 28.4$, $k = 71.6$

e. 0.007

f. $\text{H}_0 : \mu = 50$, $\text{H}_1 : \mu > 50$
Critical region is $\overline{X} > 53.63$. Reject H_0. There is evidence that the reading scores have improved.

g. Increase could have been due to other factors, e.g. second time taking test increases familiarity.